FINAL REPORT

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Team: 20

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Project Title: Enhancing Enterprise Knowledge Systems with Multi-Modal RAG

Executive Summary

Company Background and Problem Statement

Infineon Technologies manages a vast and continuously growing repository of operational knowledge stored as multi-modal documents — technical slide decks, engineering reports, and product specification sheets — that combine dense textual content with charts, workflow diagrams, and system architecture schematics. Existing enterprise search infrastructure is fundamentally text-only: it relies on Optical Character Recognition (OCR) to extract raw characters from documents but is incapable of interpreting spatial relationships, chart structures, or diagram layouts. Up to 40% of the informational content embedded within these documents is consequently invisible to current retrieval systems. Because of this, engineers spend 8-12 hours per week manually reviewing slide decks to locate specific visual data.

What We Delivered

We developed and implemented MMRAG v3, a pilot-ready Multimodal Retrieval-Augmented Generation prototype tailored for enterprise slide deck question-answering. The system replaces existing OCR-only workflows with a two-phase architecture: an offline indexing stage that combines EasyOCR text extraction with AI-generated visual captions produced by Groq Llama-4-Scout (17 billion parameters), and an online query stage that performs dual-path retrieval by fusing semantic text similarity (MiniLM, 384-dimensional vectors) with visual concept matching (CLIP ViT-B/32, 512-dimensional vectors). Answers are generated by submitting the retrieved slide image directly to the vision-language model, grounding every response in a physical visual source, and significantly reducing hallucination risk compared to text-only approaches.

Key Results

Evaluated across a curated four-tier golden dataset on the Infineon ams OSRAM acquisition deck, the optimal 0.75 text + 0.25 image retrieval configuration achieved a 96% overall score averaged across Tiers 1–3, with a 100% hit rate, 97% fact coverage, and 92% answer quality as judged by an independent LLM evaluator. Tested against a harder 36-slide Infineon quarterly report with greater retrieval ambiguity, the system achieved a 77% overall score and 92% hit rate — demonstrating strong retrieval performance but reduced generation quality under higher document complexity. These results position the system as ready for controlled internal piloting on curated document sets, with multi-slide synthesis identified as the primary capability gap to address before broader deployment. The full system runs on CPU and has been validated on a standard Windows laptop with 16 GB of RAM.

Recommendations and Caveats

We recommend that Infineon deploy MMRAG v3 as a controlled internal pilot for document querying on curated slide sets, as a step toward broader enterprise adoption. The primary implementation caveat is that initial offline indexing requires approximately one minute of processing per document page on standard CPU hardware. Planned extensions to close remaining gaps include FAISS-based approximate nearest-neighbors indexing larger databases with over 1000 slides, multi-slide answer synthesis to address the cross-document retrieval shortfall observed in the larger dataset, and cloud deployment for team-wide access.

1. Company Introduction

Infineon Technologies is a German semiconductor manufacturer and one of the ten largest chip producers globally by revenue, with operations spanning more than 40 countries and a workforce of approximately 56,000 employees. The company specializes in power semiconductors, microcontrollers, and sensor solutions across automotive, industrial, and consumer electronics. Its portfolio ranges from automotive-grade microcontrollers and silicon carbide power modules to radar sensors and MEMS microphones, reflecting the breadth of technical documentation the organization generates and maintains internally.

Internally this generates a continuous stream of multi-modal documents, product roadmap presentations, acquisition briefings, quarterly reports, and engineering specifications updated frequently and routinely queried by engineers seeking specific figures, tables, or diagrams from presentations produced months or years prior.

The business case is straightforward: if engineers currently spend 8–12 hours per week manually searching for slide decks, even a 50% reduction recovers 80–120 engineering hours per week across a 20-person team. This is a productivity gain that substantially exceeds implementation overhead.

2. Problem Definition and Motivation

2.1 The Core Technical Limitation

Standard Retrieval-Augmented Generation (RAG) systems operate on a straightforward principle: textual content is extracted from documents, converted into dense vector representations, and retrieved in response to a user query via semantic similarity search. This paradigm performs well when documents are predominantly textual but degrades significantly when applied to slide decks and technical reports where meaningful information is embedded within visual structures such as pie charts, sensor portfolio grids, system architecture flow diagrams, and annotated schematics. Two compounding failure modes emerge directly from this limitation.

2.2 Motivation

The primary motivation for this project was to close the gap between the rich informational content present in Infineon's visual documents and the severely limited capacity of existing tools to access it. A system capable of searching, retrieving, and reasoning across both textual and visual content with the same contextual fluency a human analyst brings to the task. This would eliminate the need for manual document review and transform previously inaccessible charts and diagrams into fully query-able organizational knowledge assets.

3. Methodology and Tools

3.1 System Architecture Overview

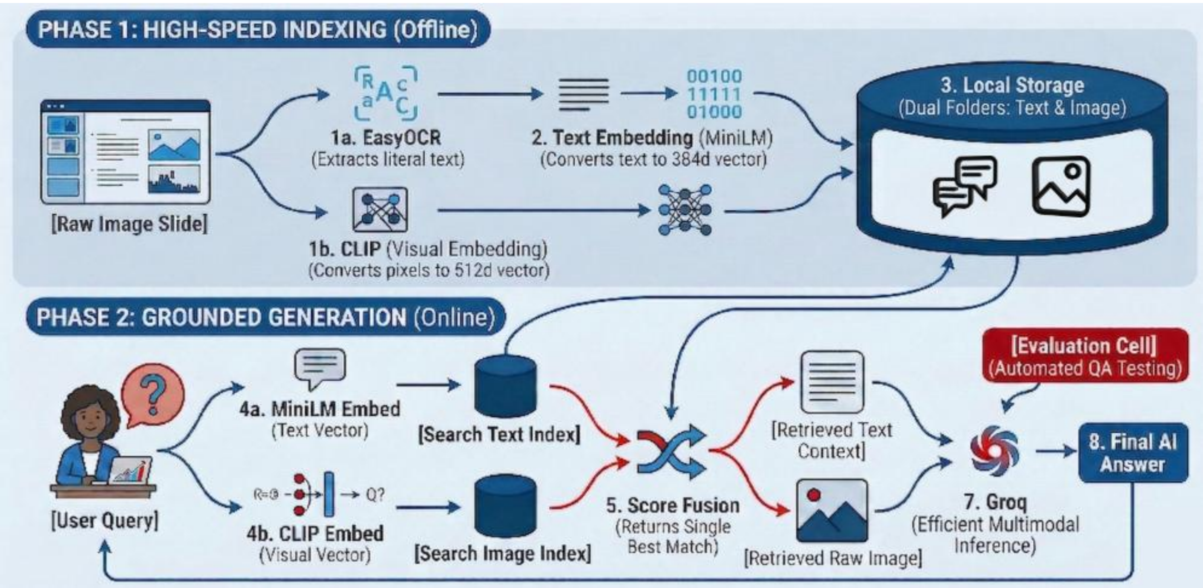

Figure 1: MMRAG v3 two-phase architecture.

MMRAG v3 uses a two-phase architecture: offline indexing for document understanding and online retrieval-generation for live user queries. This separation keeps computationally expensive visual captioning out of the query path, reducing live response latency while preserving rich multimodal document understanding.

3.2 Phase 1: Data Ingestion and Offline Indexing

When a document is uploaded, it is first rendered to a sequence of page images at 200 dots per inch using the pdf2image library. Each page image is then passed through a dual-extraction pipeline:

- Text extraction via EasyOCR: the image is scanned to extract all readable text with a minimum confidence threshold of 50%. This threshold was set empirically to suppress low-confidence noise while retaining partially rotated or degraded text that stricter thresholds would discard. One limitation is that handwriting is harder to read and therefore the confidence level is lower.
- Visual layout translation via Groq Llama-4-Scout: the raw page image is submitted to the vision-language model with a structured prompt instructing it to produce an exhaustive description of all visual elements present such as tables, diagrams, arrows, labelled axes, chart types, and spatial relationships. This step translates the visual semantics of each slide into a text representation that can be embedded and searched.

Following dual extraction, each slide is automatically classified as text-dominant or visual-dominant using two heuristics: fewer than 150 OCR words, or more than 12% numerical digit density. Visual-dominant slides use the AI caption as the primary retrieval signal; text-dominant slides use raw OCR. The combined text is then encoded into two separate vector spaces:

- Semantic text index: the all-MiniLM-L6-v2 model encodes the combined text into a 384-dimensional vector capturing semantic meaning and contextual relationships.
- Visual concept index: the page image is passed through CLIP ViT-B/32 to produce a 512-dimensional embedding in CLIP's shared text-image embedding space, capturing the visual concept of the slide independently of its textual content.

Both indices are stored as NumPy arrays alongside slide metadata and image paths in a local JSON audit log, enabling manual inspection of extraction quality for any retrieved result.

3.3 Phase 2: Dual-Path Retrieval and Grounded Generation

When a user submits a query, it is encoded simultaneously by MiniLM and CLIP. MiniLM maps the query text into the 384-dimensional semantic space; CLIP maps the same text into the 512-dimensional visual concept space. Cosine similarity scores are then computed between each query vector and every indexed slide in the corresponding vector store. The two scores for each slide are combined using a weighted linear fusion formula:

$$\text{Final Score} = 0.75 \times \text{Text Similarity} + 0.25 \times \text{Image Similarity}$$

The 0.75/0.25 weighting was determined by a component analysis study comparing retrieval configurations. Favoring text similarity prevents visually similar but factually irrelevant slides from outranking the correct result, while the 25% image component provides a meaningful rescue signal for diagram-heavy slides where OCR text is sparse or absent. Any slide that ranks highly on both paths simultaneously receives a compounded advantage and is consistently retrieved as the top result regardless of content type.

After retrieval, answer-based re-ranking compares the generated answer against each retrieved slide via word overlap, promoting the most evidentially relevant slide to the top position in the source panel for transparent verification.

The top-K slide images and their combined text context are submitted to Groq Llama-4-Scout in multi-image mode for answer generation. The model reads the actual slide image directly, not a cached summary or intermediate representation, alongside the extracted text before producing its response. This visual grounding step substantially reduces hallucination risk by anchoring generation directly to the retrieved slide image, making unsupported decisions less likely.

3.4 Deployment and User Interface

The complete pipeline runs entirely on CPU, validated on a 16 GB RAM Windows machine with integrated graphics requiring no GPU or cloud storage. Answer generation uses the Groq API for sub-second response times, and the Streamlit UI provides adjustable Top-K and weighting controls with side-by-side OCR and visual description panels for source verification.

4. Main Results

4.1 Evaluation Framework

A four-tier golden evaluation framework was designed with 16 questions (12 scored). Tiers 1–3 contribute to the overall score; Tier 4 stress tests are excluded to avoid penalizing correct refusals.

- Tier 1 — Direct Text: single-slide factual questions (acquisition price, employee counts, close dates) serving as the performance baseline.
- Tier 2 — Visual Read: chart and diagram questions entirely invisible to text-only search, directly validating the visual captioning component.
- Tier 3 — Multi-Slide Synthesis: questions requiring evidence across multiple slides, testing cross-document reasoning within a single context window.
- Tier 4 — Stress Tests: adversarial and out-of-scope queries; results recorded for transparency but excluded from the average score.

4.2 Results on Primary Dataset: Infineon ams OSRAM Acquisition Deck

The primary evaluation was conducted on the six-slide Infineon ams OSRAM acquisition briefing, selected because it spans all three content types — text-dominant, visual-dominant, and mixed-content slides — within a compact, manually verifiable document. The 0.75 text + 0.25 image configuration produced the results reported in Table 1.

Tier	Qs	Hit %	Fact %	Quality	Score	Notes

Tier 1 — Direct Text	4	100%	100%	100%	**100%**	≥ 90% target
Tier 2 — Visual Read	4	100%	100%	100%	**100%**	60–80% target
Tier 3 — Multi-Slide	4	100%	92%	75%	**89%**	40–65% target
Tier 4 — Stress Tests	4	67%	33%	38%	**54%**	Excl. from avg

Table 1: Evaluation results on the 6-slide Infineon acquisition dataset using the 0.75/0.25 configuration (12 scored questions across Tiers 1–3). Tier 4 results are reported for transparency but excluded from the 96% overall average.

Perfect scores on Tiers 1 and 2 confirm both the OCR extraction integrity and the visual captioning effectiveness — Tier 2 is particularly significant as these chart-reading questions would return null results under any text-only approach.

The Tier 3 score of 89% reflects one OK result where the generator added Pressure sensors (Industrial only) to a cross-column answer, a known failure mode as shown in Appendix A.

Table 2 confirms that the 0.75/0.25 configuration outperforms equal weighting by 12 points; higher visual weight surfaces visually similar but factually irrelevant slides.

Configuration	Overall Score	Hit Rate	Fact Coverage
0.75 Text + 0.25 Image (optimal)	**96%**	**100%**	**97%**
0.50 Text + 0.50 Image	84%	92%	86%

Table 2: Component analysis comparing retrieval weighting configurations on the 6-slide primary dataset.

4.3 Sample Question-Answer Pairs Across Tiers

Appendix B presents representative question-answer pairs across tiers, illustrating GOOD, OK, and BAD outcomes. The Tier 2 pie chart example demonstrates the system's ability to correctly read visual content that is invisible to text-only retrieval, while the Tier 3 failure from the 36-slide dataset shows how incorrect figures can arise when the required segment breakdown slide is absent from the top-K context window.

4.4 Results on Extended Dataset: 36-Slide Infineon Quarterly Report

To evaluate performance under realistic enterprise conditions, evaluation was extended to a 36-slide Infineon quarterly investor report. The larger index introduces far greater retrieval ambiguity: multiple slides share thematic vocabulary, quarterly financial tables appear across several pages with structurally similar layouts, and cross-segment comparisons require the retrieval of slides from different sections of the document. Table 3 reports results from this run (14 questions total, 12 scored).

Tier	Qs	Hit %	Fact %	Quality	Score	Notes
Tier 1 — Direct Text	4	100%	100%	88%	**96%**	≥ 90% target
Tier 2 — Visual Read	4	100%	75%	75%	**83%**	60–80% target
Tier 3 — Multi-Slide	4	75%	54%	25%	**51%**	40–65% target
Tier 4 — Stress Tests	2	100%	100%	75%	**92%**	Excl. from avg

Table 3: Evaluation results on the 36-slide Infineon quarterly report dataset. Tier 4 comprised 2 questions on this dataset; results are excluded from the 77% overall average.

Overall performance settled at 77% (Hit Rate 92%) — the correct slide was present in top-3 for almost all questions. Principal degradation is in Tier 3 (51%), where evidence split across slides not all present in the top-K window causes the generator to fill gaps with incorrect figures. Tier 2 degradation (83%) traces to CLIP's insensitivity to axis-label differences between visually near-identical charts. Both failure patterns are analyzed in Appendix A.

4.5 System Interface

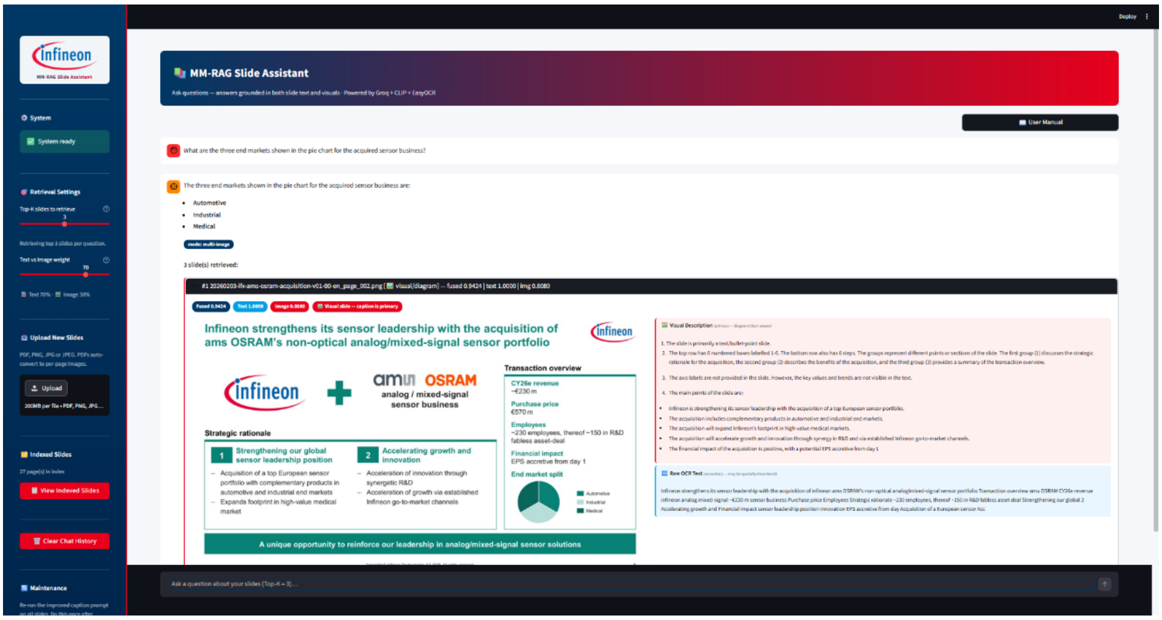

Figure 2: Retrieval panel showing grounded answers, fused scores, slide-type classification, and raw OCR for manual verification.

Figures 2 illustrates the informative pipeline steps: the retrieval panel showing fused score badges and slide-type classification, and the grounded answer with full source attribution.

5. Assumptions, Limitations, and Contributions

5.1 Assumptions

The system assumes the best answer is derivable from top-K retrieved slides and that Groq Llama-4-Scout can accurately caption each slide in isolation. These assumptions hold the document types evaluated but may degrade highly specialized diagrams or a compilation of aggregation tasks.

5.2 Limitations

Key limitations: Groq API rate limits required evaluation to be spread across multiple sessions; EasyOCR struggles with stylised or handwritten text; and the single-pass architecture cannot synthesise evidence across slides not present in the top-K window.

5.3 Contributions

Three contributions go beyond standard RAG: (1) slide-type classification routing visual and text slides to differentiated indexing pipelines; (2) answer-based re-ranking at zero inference cost; and (3) a four-tier evaluation framework with an independent LLM judge, providing a repeatable, objective benchmark.

6. Conclusion

This project addressed a concrete and measurable operational inefficiency: the inability of Infineon's existing enterprise search tools to retrieve or reason over the visual content embedded in technical slide decks and reports. The MMRAG v3 system we designed, built, and evaluated demonstrates that this gap can be meaningfully narrowed with a principled multimodal architecture at no GPU or cloud storage cost, achieving 96% overall on the curated 6-slide primary benchmark and 77% on the harder 36-slide quarterly report. Strong performance that justifies continued investment, while acknowledging that multi-slide synthesis and scale remain open to engineering challenges.

Multi-slide synthesis on larger datasets is the primary technical gap to address, with Tier 3 performance dropping from 89% to 51% as the index grew from 6 to 36 slides. FAISS-based indexing for scale and multi-turn retrieval for complementary evidence fetch are the next development priorities, upon which the architecture developed here provides a well-validated foundation for iterative progress toward a fully production-grade system.

References

Lewis, P., et al. (2020). Retrieval-Augmented Generation for knowledge-intensive NLP tasks. NeurIPS 33, 9459–9474.

Radford, A., et al. (2021). Learning transferable visual models from natural language supervision. ICML (pp. 8748–8763).

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. EMNLP 2019.

JaidedAI. (2020). EasyOCR: Ready-to-use OCR with 80+ supported languages. GitHub. https://github.com/JaidedAI/EasyOCR

Groq, Inc. (2024). Groq API documentation: Llama-4-Scout model card and inference endpoints. https://console.groq.com/docs

36-slide deck:

https://www.infineon.com/assets/row/public/documents/corporate/fy2025-q3-web-en.pdf

6-slide deck:
https://www.infineon.com/assets/row/public/documents/corporate/investors/presentations/2026/20260203-ifx-ams-osram-acquisition-v01-00-en.pdf

Appendix A: Failure Mode Analysis

Three recurring failure modes account for majority BAD and OK judgements across both evaluation datasets. Each map to a specific architectural gap.

Failure Mode 1 — Distributed evidence. The most consistent Tier 3 failure occurs when a question requires combining data points from separate slides. The system has no mechanism to identify multi-part questions and fetch complementary evidence. The segment margin improvement question in the 36-slide run (0% fact coverage, BAD) illustrates this: the quarterly summary slide was retrieved at rank 3 but the segment-level breakdown was absent, causing the generator to confabulate margin figures for three of the four segments. Fix: multi-turn retrieval that explicitly fetches complementary slides.

Failure Mode 2 — Visually similar slides. Several slides in the 36-slide dataset share an almost identical visual structure (bar charts, same palette and layout) distinguished only by axis labels. CLIP embeddings are insensitive to small axis-label differences, so visually near-identical slides are near-identical in embedding space. The revenue recovery question (Tier 2, 33% overall) failed because the annual trend chart was retrieved instead of the quarter-on-quarter overlay. Fix: axis-label-aware re-ranking post-retrieval.

Failure Mode 3 — Semantic conflation. The growth areas revenue question (Tier 3, 36-slide, 30% overall) shows the generator retrieving the correct slide (100% hit rate) but answering using segment revenue proportions rather than the separate growth-areas breakdown co-located on the same slide. Visual grounding on a single image is insufficient when that image contains multiple co-located data structures requiring disambiguation. Fix: structured data disambiguation prompting.

Appendix B: Sample Question-Answer Pairs

The table below presents four evaluated question-answer pairs across tiers, illustrating GOOD, OK, and BAD outcomes. The final row is drawn from the 36-slide dataset and shows a generation-side failure where retrieval succeeded, but the generated answer was factually incorrect.

Tier 1 Direct Text	**Q** How many employees are included in the acquisition, and how many work in R&D? **Ref** *~230 employees in total, of which ~150 are in R&D.* **A** Approximately 230 employees, of which around 150 work in R&D. *Single slide, plain text. Hit ✓ | Fact 100% | Quality GOOD | Dataset: 6-slide*	**GOOD** Overall **100%**
Tier 2 Visual Read	**Q** What are the three end markets shown in the pie chart for the acquired sensor business? **Ref** *Automotive, Industrial, and Medical. Automotive has the largest share.* **A** Automotive, Industrial, and Medical. Automotive has the largest share. *Pie chart — invisible to text-only search. Hit ✓ | Fact 100% | Quality GOOD | Dataset: 6-slide*	**GOOD** Overall **100%**
Tier 3 Multi-Slide (partial)	**Q** Which sensor types appear in both the Automotive and Industrial columns of Infineon's portfolio? **Ref** *Four types: Radar sensors, Magnetic position sensors, Gas/leakage sensors, CAPSENSE µC.* **A** Radar sensors, Magnetic position sensors, Pressure sensors, Gas/leakage sensors, CAPSENSE µC. *Correct slide retrieved but Pressure sensors (Industrial only) incorrectly included. Hit ✓ | Fact 100% | Quality OK | Dataset: 6-slide*	**OK** Overall **83%**
Tier 3 Multi-Slide (fail)	**Q** Which segment improved its margin the most from Q3 FY24 to Q3 FY25? **Ref** *Power & Sensor Systems, from 11.7% to 18.8% (+7.1 pp).* **A** Automotive segment. Margin improved from 19.6% to 19.8%, an increase of 0.2 percentage points. *Wrong slide retrieved (quarterly summary, not segment breakdown). Generator hallucinated figures. Hit ✓ (rank #3) | Fact 0% | Quality BAD | Dataset: 36-slide*	**BAD** Overall **33%**

Table B1: Representative Q&A pairs across tiers and datasets. The BAD row is from the 36-slide evaluation.